

TidyTuesday Week 32

Pranay Gundam

Saturday 8th August, 2026

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1 Weekly Summary

AI-Generated Content

The summary below was written by Claude (Anthropic) from that week's regression outputs and series descriptions. It has not been reviewed or fact-checked by a human, and should be read as an automated, informal recap – not analysis.

Welcome back to the weekly ritual where a computer flings two unrelated economic time series at each other and I pretend to be surprised by the results. This week's grab bag included Bitcoin, a discontinued San Francisco tech index, Italian central bank foreign exchange purchases, delinquency rates on real estate loans, NAFTA recession flags, U.S. exports to "NICS," a mouthful of a consumer price index, and, for continental flavor, French dwelling construction starts. Truly the theme was "no theme," which is exactly how we like it.

Monday kicked off strong with Bitcoin versus the San Francisco Tech Pulse, and the raw regression came in swinging with an R-squared of about 0.70 and a p-value so small it needs scientific notation to feel loved. Naturally I got excited, then remembered these are two things that both went up over the years, which is the correlation equivalent of noticing that both my houseplant and my nephew got taller. But here's the mildly interesting part: after detrending, it didn't fully collapse. The relationship stayed significant, just weaker, with R-squared sliding down to a humbler 0.14. So while I am legally and morally obligated to tell you this proves nothing, it's one of the rare weeks where a link survived having its shared drift surgically removed. Put a gold star on it and move on.

The rest of the week was a parade of duds, which honestly is the more honest outcome. Tuesday's Philadelphia capital expenditures diffusion index versus real user cost of money posted a p-value of 0.89, meaning "no." Wednesday's Italian foreign exchange purchases looked mildly significant against loan delinquency rates in the raw version, then evaporated the second we detrended it, p-value strolling up to 0.35 without a care in the world. That, friends, is the textbook trend-mirage this project exists to point and laugh at. Wednesday's NAFTA recession indicator versus current account balance never even bothered to look interesting, and Thursday's U.S. exports versus state-and-local government spending share flipped its coefficient sign between raw and detrended, which is the statistical version of a witness changing their story on the stand.

The week's closing weirdness came Friday, when a consumer price index for nondurables cozied up to French residential construction starts. The raw regression shrugged (p-value 0.12), but the detrended version squeaked under the significance line at 0.04, insisting on a positive relationship. Do American nondurable prices secretly whisper sweet nothings to French home-builders? Almost certainly not, and with an R-squared of 0.08 it's a whisper at best. As always, the standing disclaimer applies with full force: these are randomly paired series, no causation is claimed, no theory is endorsed, and spurious correlation is the house specialty rather than a bug. See you next week for more numbers that mean absolutely nothing together.

2 Date: 2026-08-03

Series ID: CBBTCUSD

This series is titled Coinbase Bitcoin and has a frequency of Daily, 7-Day. The units are U.S. Dollars and the seasonal adjustment is Not Seasonally Adjusted. The observation start date is 2014-12-01 and the observation end date is 2026-08-02. The popularity of this series is 75.

Series ID: SFTPINDM114SFRBSF

This series is titled San Francisco Tech Pulse (DISCONTINUED) and has a frequency of Monthly. The units are Index Jan 2000=100 and the seasonal adjustment is Seasonally Adjusted. The observation start date is 1971-04-01 and the observation end date is 2020-03-01. The popularity of this series is 10.

2.1 Raw Regression

Regresses the two series against each other directly. A significant result here can come just as easily from both series sharing a long-run trend as from any real relationship.

Dep. Variable:	value_fred_SFTPINDM114SFRBSF	R-squared:	0.695
Model:	OLS	Adj. R-squared:	0.690
Method:	Least Squares	F-statistic:	139.1
Date:	Mon, 03 Aug 2026	Prob (F-statistic):	2.21e-17
Time:	14:09:51	Log-Likelihood:	-185.92
No. Observations:	63	AIC:	375.8
Df Residuals:	61	BIC:	380.1
Df Model:	1		
Covariance Type:	nonrobust		

	coef	std err	t	P> t	[0.025	0.975]
const	84.9060	0.863	98.348	0.000	83.180	86.632
value_fred_CBBTCUSD	0.0018	0.000	11.796	0.000	0.002	0.002

Omnibus:	9.975	Durbin-Watson:	0.305
Prob(Omnibus):	0.007	Jarque-Bera (JB):	10.825
Skew:	0.715	Prob(JB):	0.00446
Kurtosis:	4.442	Cond. No.	8.11e+03

Notes:

[1] Standard Errors assume that the covariance matrix of the errors is correctly specified.

[2] The condition number is large, 8.11e+03. This might indicate that there are strong multicollinearity or other numerical problems.

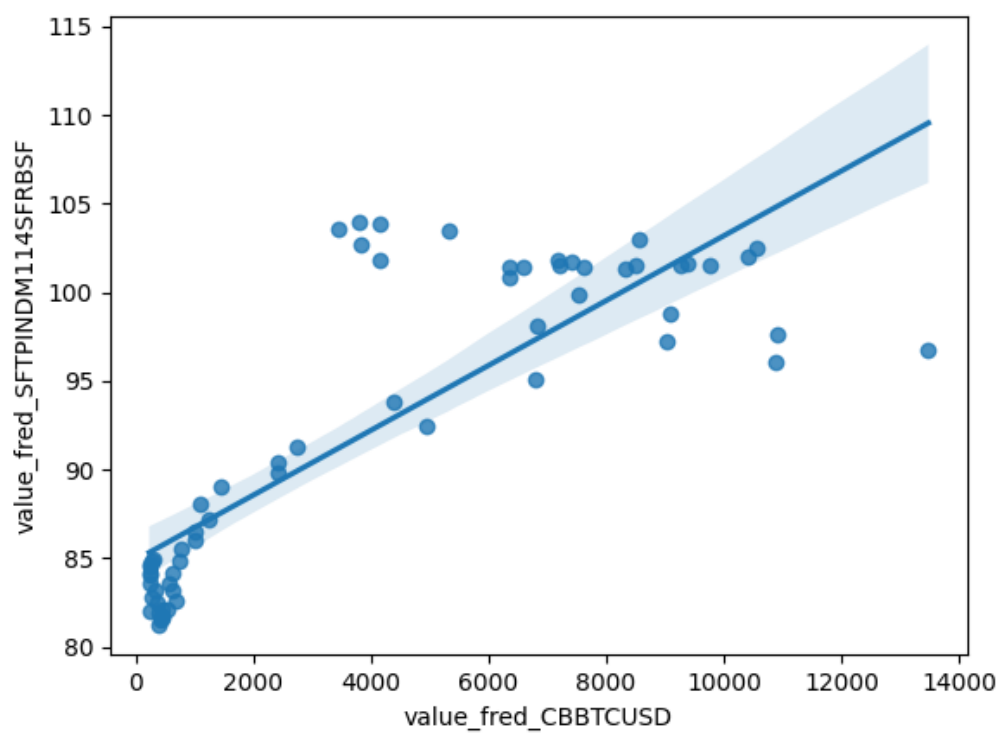


Figure 1: Raw Regression Plot for 2026-08-03

2.2 Detrended Regression

Regresses the residuals of each series after removing its own linear time trend, so a shared trend can no longer drive the result on its own. A relationship that stays significant here is better evidence of a genuine link between the two series.

Dep. Variable:	value_fred_SFTPINDM114SFRBSF_detrended	R-squared:	0.137
Model:	OLS	Adj. R-squared:	0.123
Method:	Least Squares	F-statistic:	9.672
Date:	Mon, 03 Aug 2026	Prob (F-statistic):	0.00284
Time:	14:09:51	Log-Likelihood:	-150.74
No. Observations:	63	AIC:	305.5
Df Residuals:	61	BIC:	309.8
Df Model:	1		
Covariance Type:	nonrobust		

	coef	std err	t	P> t	[0.025	0.975]
const	3.857e-12	0.339	1.14e-11	1.000	-0.678	0.678
value_fred_CBBTCUSD_detrended	0.0005	0.000	3.110	0.003	0.000	0.001

Omnibus:	23.048	Durbin-Watson:	0.095
Prob(Omnibus):	0.000	Jarque-Bera (JB):	4.918
Skew:	0.258	Prob(JB):	0.0855
Kurtosis:	1.732	Cond. No.	2.25e+03

Notes:

- [1] Standard Errors assume that the covariance matrix of the errors is correctly specified.
- [2] The condition number is large, 2.25e+03. This might indicate that there are strong multicollinearity or other numerical problems.

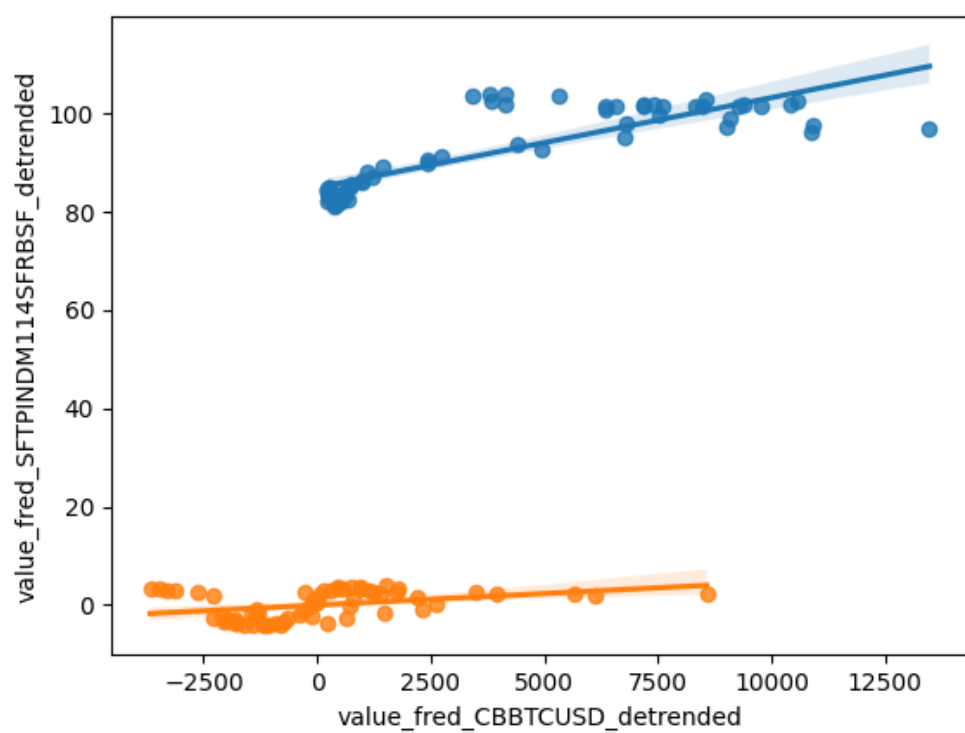


Figure 2: Detrended Regression Plot for 2026-08-03

3 Date: 2026-08-04

Series ID: CPBNDIF066MSFRBPHI

This series is titled Current Physical Plant Capital Expenditures; Diffusion Index for Federal Reserve District 3: Philadelphia and has a frequency of Monthly. The units are Index and the seasonal adjustment is Seasonally Adjusted. The observation start date is 2011-03-01 and the observation end date is 2026-07-01. The popularity of this series is 1.

Series ID: OCM2MP

This series is titled Real User Cost Index of MSI-M2M (preferred) and has a frequency of Monthly. The units are Percent and the seasonal adjustment is Not Seasonally Adjusted. The observation start date is 1967-01-01 and the observation end date is 2013-12-01. The popularity of this series is 4.

3.1 Raw Regression

Regresses the two series against each other directly. A significant result here can come just as easily from both series sharing a long-run trend as from any real relationship.

Dep. Variable:	value_fred_OCM2MP	R-squared:	0.001
Model:	OLS	Adj. R-squared:	-0.031
Method:	Least Squares	F-statistic:	0.01870
Date:	Tue, 04 Aug 2026	Prob (F-statistic):	0.892
Time:	13:31:41	Log-Likelihood:	184.86
No. Observations:	34	AIC:	-365.7
Df Residuals:	32	BIC:	-362.7
Df Model:	1		
Covariance Type:	nonrobust		

	coef	std err	t	P> t	[0.025	0.975]
const	0.0155	0.000	36.337	0.000	0.015	0.016
value_fred_CPBNDIF066MSFRBPHI	3.385e-06	2.47e-05	0.137	0.892	-4.7e-05	5.38e-05

Omnibus:	9.983	Durbin-Watson:	0.076
Prob(Omnibus):	0.007	Jarque-Bera (JB):	2.548
Skew:	0.142	Prob(JB):	0.280
Kurtosis:	1.689	Cond. No.	39.7

Notes:

[1] Standard Errors assume that the covariance matrix of the errors is correctly specified.

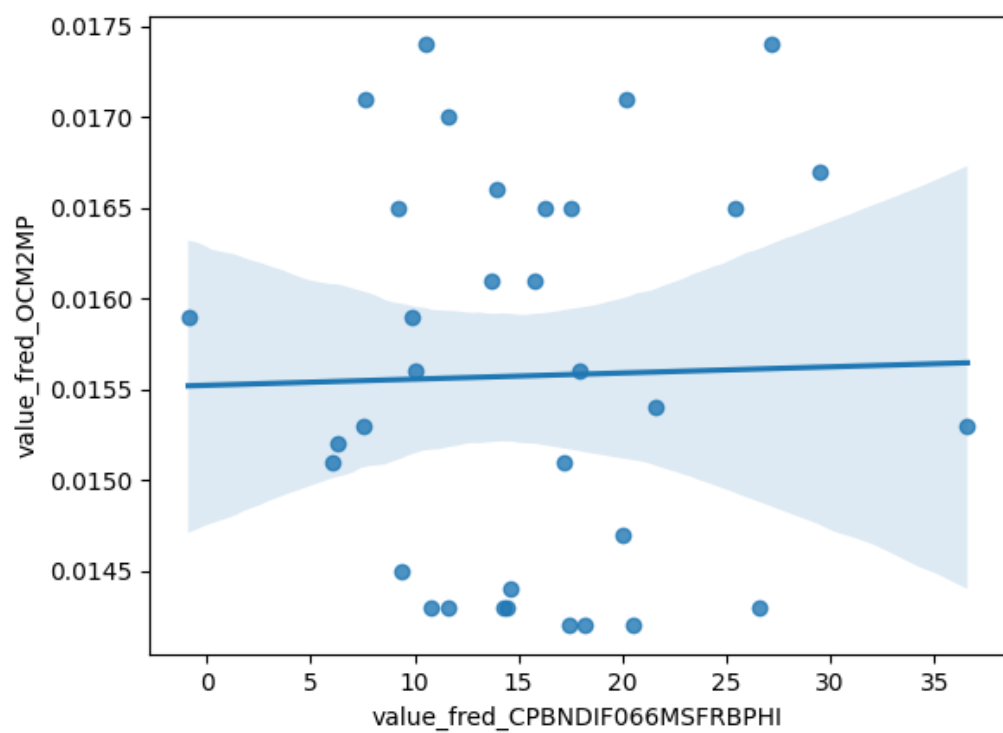


Figure 3: Raw Regression Plot for 2026-08-04

3.2 Detrended Regression

Regresses the residuals of each series after removing its own linear time trend, so a shared trend can no longer drive the result on its own. A relationship that stays significant here is better evidence of a genuine link between the two series.

Dep. Variable:	value_fred_OCM2MP_detrended	R-squared:	0.000
Model:	OLS	Adj. R-squared:	-0.031
Method:	Least Squares	F-statistic:	0.006047
Date:	Tue, 04 Aug 2026	Prob (F-statistic):	0.939
Time:	13:31:41	Log-Likelihood:	193.17
No. Observations:	34	AIC:	-382.3
Df Residuals:	32	BIC:	-379.3
Df Model:	1		
Covariance Type:	nonrobust		

	coef	std err	t	P> t	[0.025	0.975]
const	4.858e-15	0.000	3.33e-11	1.000	-0.000	0.000
value_fred_CPBNDIF066MSFRBPHI_detrended	1.508e-06	1.94e-05	0.078	0.939	-3.8e-05	4.1e-05

Omnibus:	10.150	Durbin-Watson:	0.122
Prob(Omnibus):	0.006	Jarque-Bera (JB):	2.467
Skew:	0.016	Prob(JB):	0.291
Kurtosis:	1.681	Cond. No.	7.52

Notes:

[1] Standard Errors assume that the covariance matrix of the errors is correctly specified.

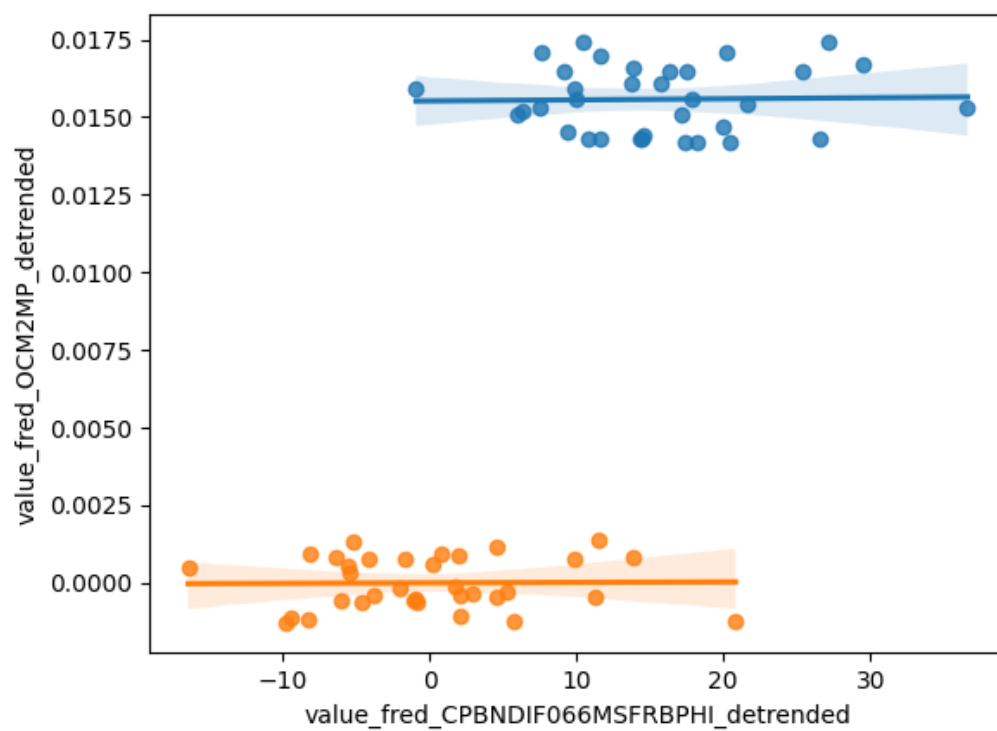


Figure 4: Detrended Regression Plot for 2026-08-04

4 Date: 2026-08-05

Series ID: ITINTDRES

This series is titled Italian Intervention: Banca d'Italia Purchases of Foreign Exchange (Millions of ECU's, Euros after 1999) (DISCONTINUED) and has a frequency of Daily, 7-Day. The units are Millions of ECU's, Euros after 1999 and the seasonal adjustment is Not Seasonally Adjusted. The observation start date is 1988-01-01 and the observation end date is 1998-12-31. The popularity of this series is 2.

Series ID: DRSRET100N

This series is titled Delinquency Rate on Loans Secured by Real Estate, Banks Ranked 1st to 100th Largest in Size by Assets and has a frequency of Quarterly, End of Period. The units are Percent and the seasonal adjustment is Not Seasonally Adjusted. The observation start date is 1987-01-01 and the observation end date is 2026-01-01. The popularity of this series is 1.

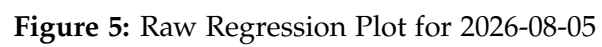
4.1 Raw Regression

Regresses the two series against each other directly. A significant result here can come just as easily from both series sharing a long-run trend as from any real relationship.

Dep. Variable:	value_fred_DRSRET100N	R-squared:	0.095			
Model:	OLS	Adj. R-squared:	0.074			
Method:	Least Squares	F-statistic:	4.433			
Date:	Wed, 05 Aug 2026	Prob (F-statistic):	0.0413			
Time:	13:32:37	Log-Likelihood:	-100.49			
No. Observations:	44	AIC:	205.0			
Df Residuals:	42	BIC:	208.5			
Df Model:	1					
Covariance Type:	nonrobust					
	coef	std err	t	P> t	[0.025	0.975]
const	5.6231	0.377	14.905	0.000	4.862	6.384
value_fred_ITINTDRES	-0.0081	0.004	-2.105	0.041	-0.016	-0.000
	Omnibus:	5.847	Durbin-Watson:	0.227		
	Prob(Omnibus):	0.054	Jarque-Bera (JB):	2.476		
	Skew:	0.255	Prob(JB):	0.290		
	Kurtosis:	1.956	Cond. No.	101.		

Notes:

[1] Standard Errors assume that the covariance matrix of the errors is correctly specified.



4.2 Detrended Regression

Regresses the residuals of each series after removing its own linear time trend, so a shared trend can no longer drive the result on its own. A relationship that stays significant here is better evidence of a genuine link between the two series.

Dep. Variable:	value_fred_DRSRET100N_detrended	R-squared:	0.021
Model:	OLS	Adj. R-squared:	-0.002
Method:	Least Squares	F-statistic:	0.9112
Date:	Wed, 05 Aug 2026	Prob (F-statistic):	0.345
Time:	13:32:38	Log-Likelihood:	-87.537
No. Observations:	44	AIC:	179.1
Df Residuals:	42	BIC:	182.6
Df Model:	1		
Covariance Type:	nonrobust		

	coef	std err	t	P> t	[0.025	0.975]
const	-5.314e-13	0.273	-1.95e-12	1.000	-0.551	0.551
value_fred_ITINTDRES_detrended	-0.0029	0.003	-0.955	0.345	-0.009	0.003

Omnibus:	3.670	Durbin-Watson:	0.127
Prob(Omnibus):	0.160	Jarque-Bera (JB):	3.038
Skew:	0.532	Prob(JB):	0.219
Kurtosis:	2.275	Cond. No.	90.8

Notes:

[1] Standard Errors assume that the covariance matrix of the errors is correctly specified.

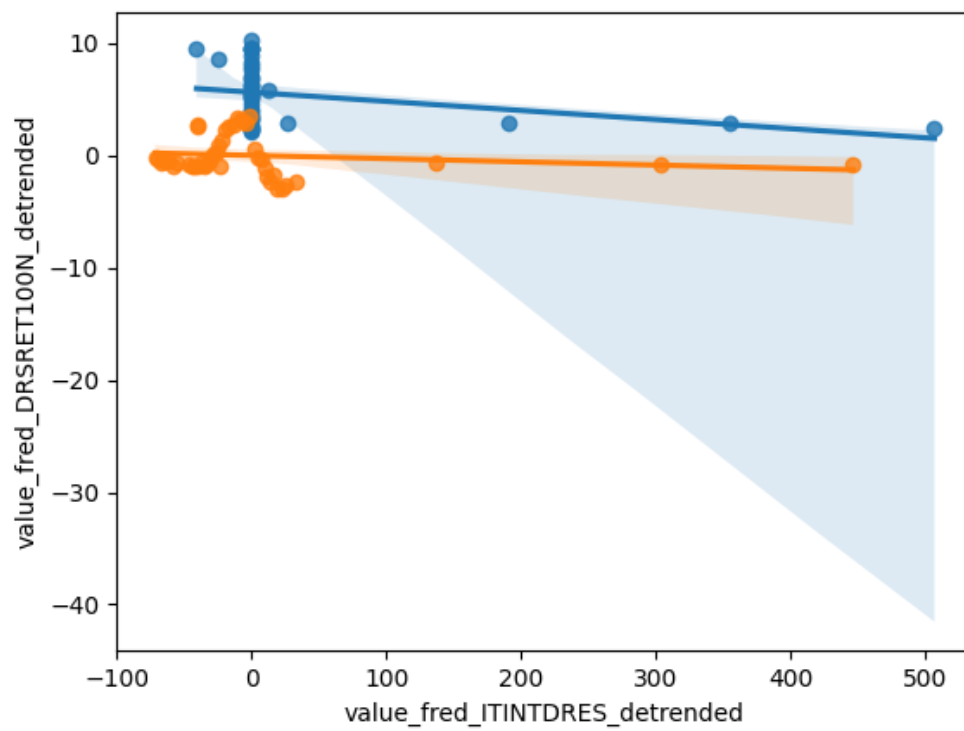


Figure 6: Detrended Regression Plot for 2026-08-05

5 Date: 2026-08-06

Series ID: NAFTARECP

This series is titled OECD based Recession Indicators for NAFTA Area from the Peak through the Period preceding the Trough (DISCONTINUED) and has a frequency of Monthly. The units are +1 or 0 and the seasonal adjustment is Not Seasonally Adjusted. The observation start date is 1947-03-01 and the observation end date is 2022-08-01. The popularity of this series is 1.

Series ID: BOPBCAA

This series is titled Balance on Current Account (DISCONTINUED) and has a frequency of Annual. The units are Billions of Dollars and the seasonal adjustment is Not Seasonally Adjusted. The observation start date is 1960-01-01 and the observation end date is 2013-01-01. The popularity of this series is 6.

5.1 Raw Regression

Regresses the two series against each other directly. A significant result here can come just as easily from both series sharing a long-run trend as from any real relationship.

Dep. Variable:	value_fred_BOPBCAA	R-squared:	0.005			
Model:	OLS	Adj. R-squared:	-0.014			
Method:	Least Squares	F-statistic:	0.2814			
Date:	Thu, 06 Aug 2026	Prob (F-statistic):	0.598			
Time:	13:27:58	Log-Likelihood:	-371.18			
No. Observations:	54	AIC:	746.4			
Df Residuals:	52	BIC:	750.3			
Df Model:	1					
Covariance Type:	nonrobust					
	coef	std err	t	P> t 	[0.025	0.975]
const	-188.9963	40.873	-4.624	0.000	-271.014	-106.979
value_fred_NAFTARECP	35.6291	67.161	0.531	0.598	-99.139	170.398
Omnibus:	12.191	Durbin-Watson:	0.065			
Prob(Omnibus):	0.002	Jarque-Bera (JB):	13.569			
Skew:	-1.223	Prob(JB):	0.00113			
Kurtosis:	3.210	Cond. No.	2.43			

Notes:

[1] Standard Errors assume that the covariance matrix of the errors is correctly specified.

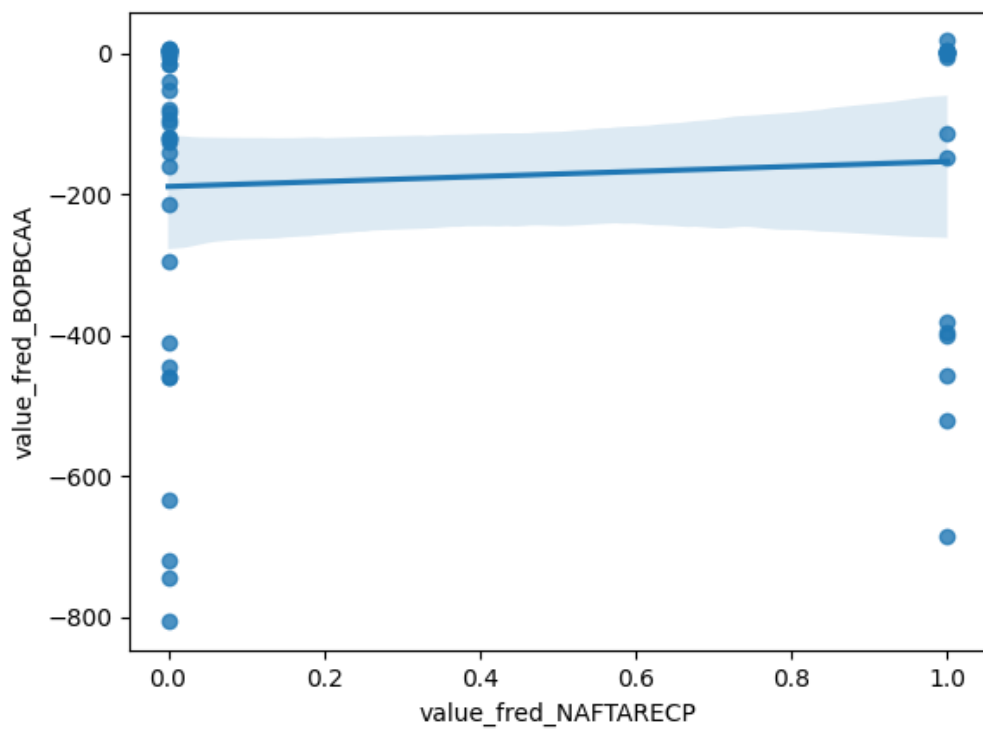


Figure 7: Raw Regression Plot for 2026-08-06

5.2 Detrended Regression

Regresses the residuals of each series after removing its own linear time trend, so a shared trend can no longer drive the result on its own. A relationship that stays significant here is better evidence of a genuine link between the two series.

Dep. Variable:	value_fred_BOPBCAA_detrended	R-squared:	0.000
Model:	OLS	Adj. R-squared:	-0.019
Method:	Least Squares	F-statistic:	0.02165
Date:	Thu, 06 Aug 2026	Prob (F-statistic):	0.884
Time:	13:27:58	Log-Likelihood:	-340.93
No. Observations:	54	AIC:	685.9
Df Residuals:	52	BIC:	689.8
Df Model:	1		
Covariance Type:	nonrobust		

	coef	std err	t	P> t	[0.025	0.975]
const	7.162e-12	18.522	3.87e-13	1.000	-37.167	37.167
value_fred_NAFTARECP_detrended	-5.6739	38.562	-0.147	0.884	-83.055	71.707

Omnibus:	8.580	Durbin-Watson:	0.207
Prob(Omnibus):	0.014	Jarque-Bera (JB):	7.858
Skew:	-0.877	Prob(JB):	0.0197
Kurtosis:	3.646	Cond. No.	2.08

Notes:

[1] Standard Errors assume that the covariance matrix of the errors is correctly specified.

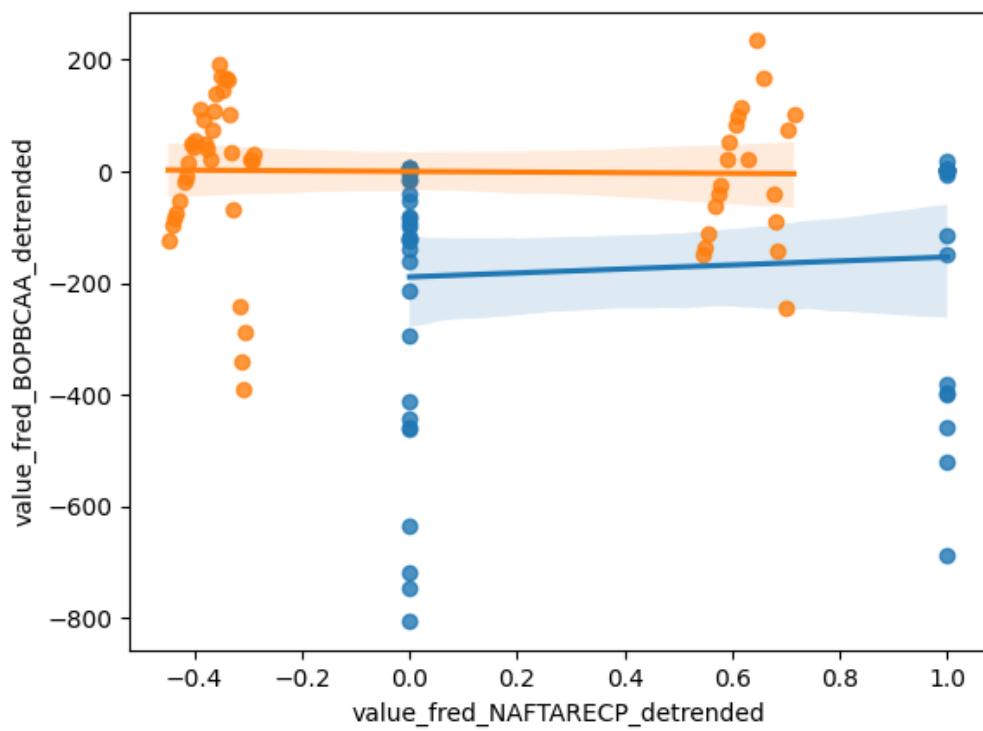


Figure 8: Detrended Regression Plot for 2026-08-06

6 Date: 2026-08-07

Series ID: EXP0008

This series is titled U.S. Exports of Goods by F.A.S. Basis to NICS and has a frequency of Monthly. The units are Millions of Dollars and the seasonal adjustment is Not Seasonally Adjusted. The observation start date is 1985-01-01 and the observation end date is 2012-12-01. The popularity of this series is 1.

Series ID: A829RE1Q156NBEA

This series is titled Shares of gross domestic product: Government consumption expenditures and gross investment: State and local and has a frequency of Quarterly. The units are Percent and the seasonal adjustment is Not Seasonally Adjusted. The observation start date is 1947-01-01 and the observation end date is 2026-04-01. The popularity of this series is 3.

6.1 Raw Regression

Regresses the two series against each other directly. A significant result here can come just as easily from both series sharing a long-run trend as from any real relationship.

Dep. Variable:	value_fred_A829RE1Q156NBEA	R-squared:	0.417
Model:	OLS	Adj. R-squared:	0.411
Method:	Least Squares	F-statistic:	78.58
Date:	Fri, 07 Aug 2026	Prob (F-statistic):	1.55e-14
Time:	12:35:32	Log-Likelihood:	-58.542
No. Observations:	112	AIC:	121.1
Df Residuals:	110	BIC:	126.5
Df Model:	1		
Covariance Type:	nonrobust		

	coef	std err	t	P> t	[0.025	0.975]
const	10.8148	0.095	113.844	0.000	10.627	11.003
value_fred_EXP0008	0.0001	1.49e-05	8.864	0.000	0.000	0.000

Omnibus:	7.279	Durbin-Watson:	0.127
Prob(Omnibus):	0.026	Jarque-Bera (JB):	6.829
Skew:	0.541	Prob(JB):	0.0329
Kurtosis:	3.540	Cond. No.	1.56e+04

Notes:

[1] Standard Errors assume that the covariance matrix of the errors is correctly specified.

[2] The condition number is large, 1.56e+04. This might indicate that there are strong multicollinearity or other numerical problems.

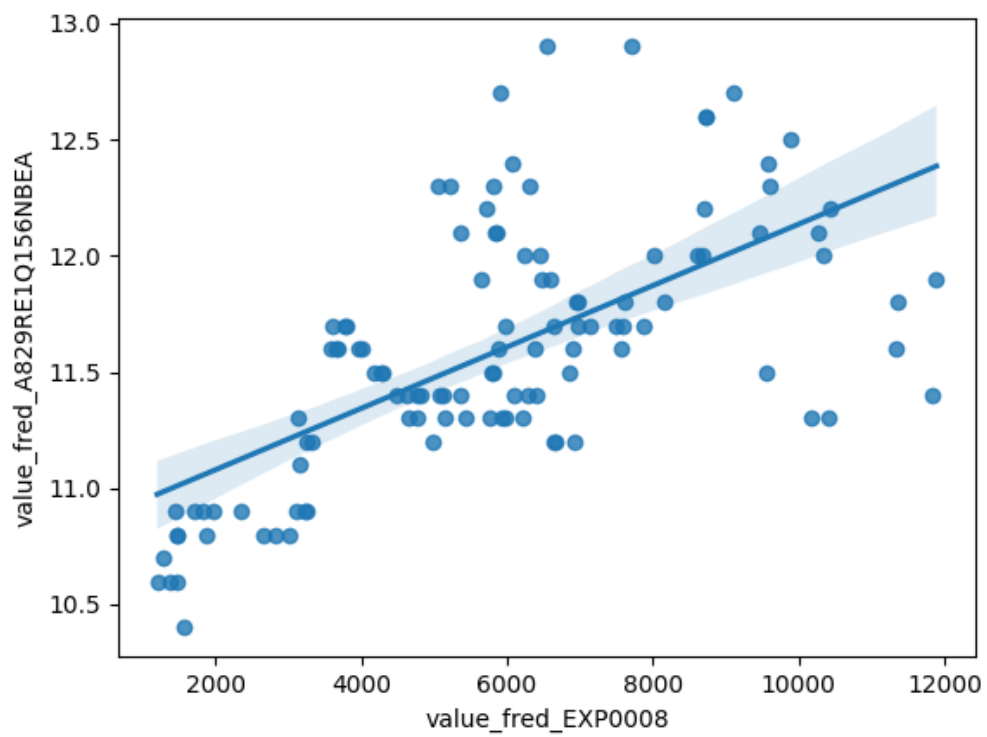


Figure 9: Raw Regression Plot for 2026-08-07

6.2 Detrended Regression

Regresses the residuals of each series after removing its own linear time trend, so a shared trend can no longer drive the result on its own. A relationship that stays significant here is better evidence of a genuine link between the two series.

Dep. Variable:	value_fred_A829RE1Q156NBEA_detrended	R-squared:	0.150
Model:	OLS	Adj. R-squared:	0.142
Method:	Least Squares	F-statistic:	19.42
Date:	Fri, 07 Aug 2026	Prob (F-statistic):	2.45e-05
Time:	12:35:32	Log-Likelihood:	-28.396
No. Observations:	112	AIC:	60.79
Df Residuals:	110	BIC:	66.23
Df Model:	1		
Covariance Type:	nonrobust		

	coef	std err	t	P> t	[0.025	0.975]
const	5.223e-13	0.030	1.76e-11	1.000	-0.059	0.059
value_fred_EXP0008_detrended	-0.0001	3.37e-05	-4.407	0.000	-0.000	-8.16e-05

Omnibus:	3.171	Durbin-Watson:	0.177
Prob(Omnibus):	0.205	Jarque-Bera (JB):	2.615
Skew:	-0.353	Prob(JB):	0.271
Kurtosis:	3.250	Cond. No.	883.

Notes:

[1] Standard Errors assume that the covariance matrix of the errors is correctly specified.

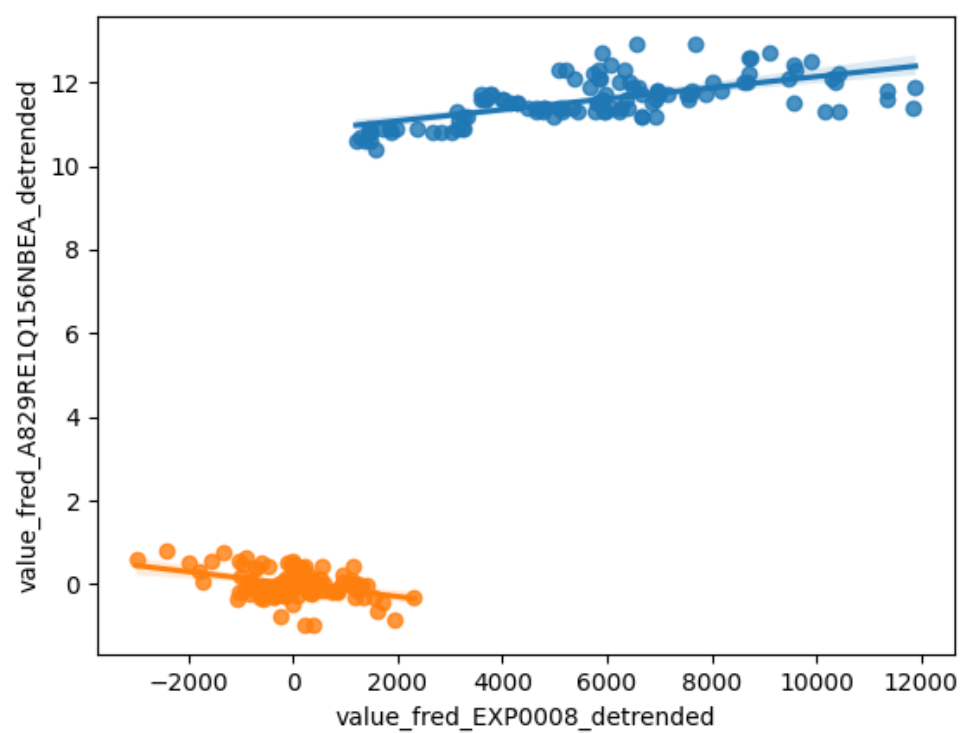


Figure 10: Detrended Regression Plot for 2026-08-07

7 Date: 2026-08-08

Series ID: CWUR0000SANL113

This series is titled Consumer Price Index for All Urban Wage Earners and Clerical Workers: Non-durables Less Food, Beverages, and Apparel in U.S. City Average and has a frequency of Monthly. The units are Index 1982-1984=100 and the seasonal adjustment is Not Seasonally Adjusted. The observation start date is 1967-01-01 and the observation end date is 2026-06-01. The popularity of this series is 1.

Series ID: WSCNDW01FRA489N

This series is titled Work Started: Construction: Dwellings and Residential Buildings: Total for France and has a frequency of Annual. The units are Number, Monthly level and the seasonal adjustment is Not Seasonally Adjusted. The observation start date is 1960-01-01 and the observation end date is 2017-01-01. The popularity of this series is 1.

7.1 Raw Regression

Regresses the two series against each other directly. A significant result here can come just as easily from both series sharing a long-run trend as from any real relationship.

Dep. Variable:	value_fred_WSCNDW01FRA489N	R-squared:	0.048
Model:	OLS	Adj. R-squared:	0.028
Method:	Least Squares	F-statistic:	2.448
Date:	Sat, 08 Aug 2026	Prob (F-statistic):	0.124
Time:	12:20:51	Log-Likelihood:	-517.37
No. Observations:	51	AIC:	1039.
Df Residuals:	49	BIC:	1043.
Df Model:	1		
Covariance Type:	nonrobust		

	coef	std err	t	P> t	[0.025	0.975]
const	3.398e+04	1785.119	19.035	0.000	3.04e+04	3.76e+04
value_fred_CWUR0000SANL113	-18.2426	11.660	-1.565	0.124	-41.674	5.189

Omnibus:	9.260	Durbin-Watson:	0.207
Prob(Omnibus):	0.010	Jarque-Bera (JB):	3.089
Skew:	0.227	Prob(JB):	0.213
Kurtosis:	1.883	Cond. No.	311.

Notes:

[1] Standard Errors assume that the covariance matrix of the errors is correctly specified.

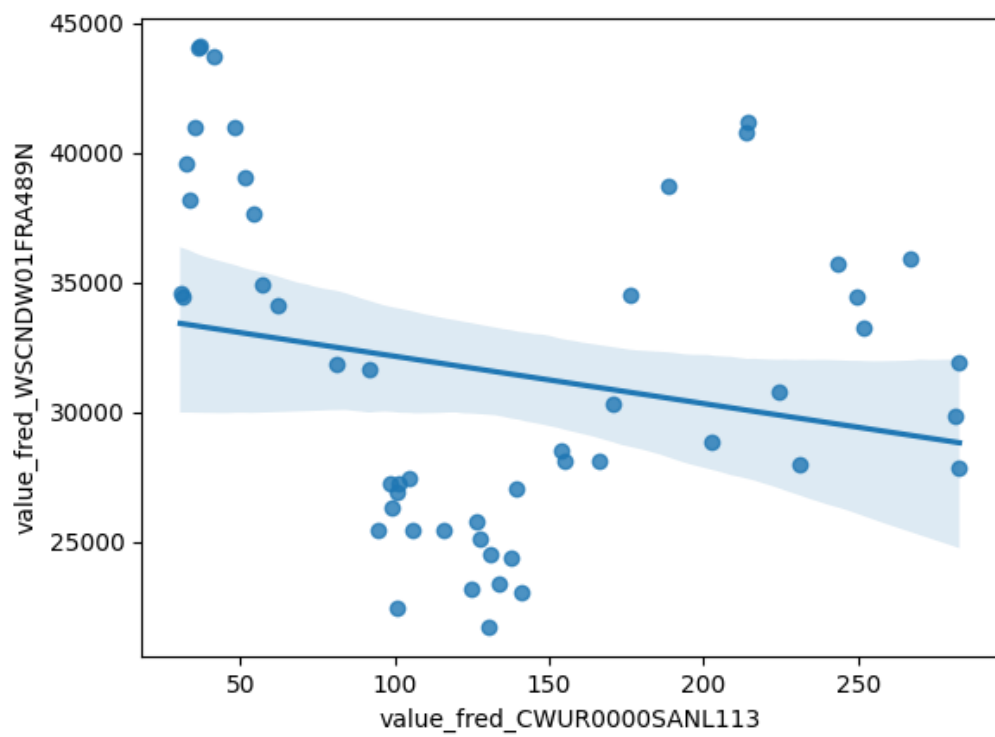


Figure 11: Raw Regression Plot for 2026-08-08

7.2 Detrended Regression

Regresses the residuals of each series after removing its own linear time trend, so a shared trend can no longer drive the result on its own. A relationship that stays significant here is better evidence of a genuine link between the two series.

Dep. Variable:	value_fred_WSCNDW01FRA489N_detrended	R-squared:	0.082
Model:	OLS	Adj. R-squared:	0.064
Method:	Least Squares	F-statistic:	4.405
Date:	Sat, 08 Aug 2026	Prob (F-statistic):	0.0410
Time:	12:20:51	Log-Likelihood:	-514.05
No. Observations:	51	AIC:	1032.
Df Residuals:	49	BIC:	1036.
Df Model:	1		
Covariance Type:	nonrobust		

	coef	std err	t	P> t	[0.025	0.975]
const	-5.457e-11	824.380	-6.62e-14	1.000	-1656.654	1656.654
value_fred_CWUR0000SANL113_detrended	90.3527	43.049	2.099	0.041	3.843	176.863

Omnibus:	4.999	Durbin-Watson:	0.260
Prob(Omnibus):	0.082	Jarque-Bera (JB):	2.635
Skew:	0.308	Prob(JB):	0.268
Kurtosis:	2.073	Cond. No.	19.1

Notes:

[1] Standard Errors assume that the covariance matrix of the errors is correctly specified.

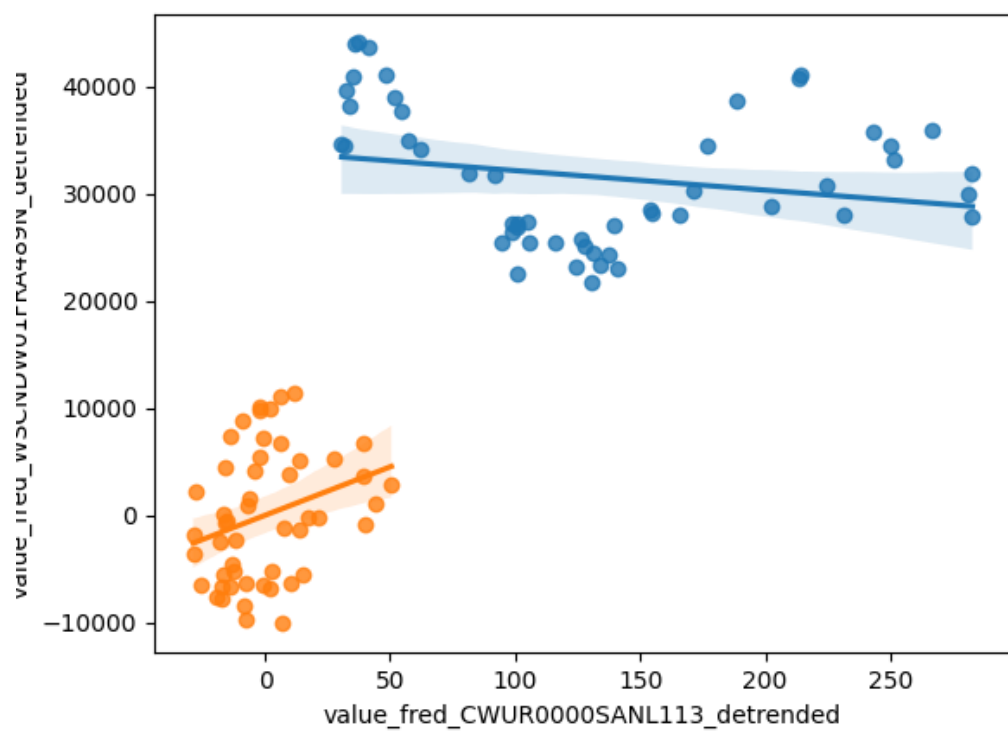


Figure 12: Detrended Regression Plot for 2026-08-08